

**MODULO 9 CLASSIFICATION OF PRIME PAIRS  $(p, p+g)$ :  
AN EMPIRICAL VERIFICATION OF THE HARDY–LITTLEWOOD  
CONJECTURE FOR  $g = 18k$**

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**ABSTRACT.** The Hardy–Littlewood conjecture for prime pairs  $(p, p+g)$  predicts the asymptotic density  $\pi(x; g) \sim H(g) x / (\log x)^2$ , where  $H(g) = 2C_2 \prod_{p|g, p>2} (p-1)/(p-2)$  is the singular series and  $C_2$  is the twin-prime constant. For  $g \equiv 0 \pmod{9}$ , classifying pairs by the digital root  $\text{dr}(p) = p \bmod 9$  partitions the population into six admissible classes  $\{1, 2, 4, 5, 7, 8\}$ , which Hardy–Littlewood in arithmetic progressions predicts to be equidistributed. We test this prediction on three fronts: (i) a control case with  $g = 2$  (twin primes, three admissible classes), where the asymptotic fractions  $1/3$  are reached within  $\pm 0.025$  percentage points up to  $x = 10^9$ , and the published aggregate counts of Oliveira e Silva (OEIS A007508) bring the agreement with  $2C_2 \text{li}_2(x)$  down to  $|\rho - 1| \leq 4 \cdot 10^{-7}$  at  $x = 10^{14}$  and  $|\rho - 1| \leq 3 \cdot 10^{-8}$  at  $x = 10^{18}$ ; (ii) the family  $g = 18k$  with  $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$ , where equidistribution over six classes holds within  $\pm 0.14$  pp and the total per-class density agrees with  $H(g)/6$  within  $\pm 0.8\%$  at  $x = 10^7$ ; (iii) the ratio  $\pi(x; g) / (H(g) \text{li}_2(x))$  converges to 1 within  $\pm 0.15\%$  at  $x = 5 \cdot 10^7$  for all tested  $k$ , including  $g = 216$  ( $k = 12$ ), whose constant  $H(g) = 4C_2$  is the smallest in the sample. The observed deviation magnitude  $|\rho - 1| \lesssim 10^{-3}$  for  $g = 18k$  at  $x = 5 \cdot 10^7$  and the empirical decay  $|\rho_{\text{glob}}(x) - 1| \asymp x^{-1/2}$  for  $g = 2$  at  $x \leq 10^{18}$  are both consistent with a square-root error term  $X^{-1/2} \log X$ , which matches the function-field analogues established unconditionally by Keating–Rudnick and Sawin–Shusterman. The results confirm Hardy–Littlewood in arithmetic progressions quantitatively within the tested regime, and show that the dr-mod-9 classification is a sensitive empirical probe that does *not* reveal deviations from the Hardy–Littlewood prediction.

## 1. INTRODUCTION

The Hardy–Littlewood conjecture on prime pairs  $(p, p+g)$ , formulated in [1], predicts that the number of such pairs with  $p \leq x$  satisfies the asymptotic

$$(1) \quad \pi(x; g) \sim H(g) \frac{x}{(\log x)^2}, \quad x \rightarrow \infty,$$

where the *Hardy–Littlewood singular series* is

$$(2) \quad H(g) = 2C_2 \prod_{\substack{p|g \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \approx 0.6601618158.$$

The conjecture is known unconditionally only for  $g = 0$  (the prime number theorem) and for no positive value of  $g$ , although deep results such as Maynard’s [4] and the Polymath 8b project [5] provide bounded gaps infinitely often. Quantitative versions restricted to arithmetic progressions are discussed in [2, 3].

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The present work addresses an operational question: *how well does formula (1)–(2) describe the distribution of prime pairs in computationally accessible regimes*, and in particular how the population partitions across residue classes modulo 9.

For every  $g \equiv 0 \pmod{9}$  (and in particular  $g = 18k$ ), the divisibility  $9 \mid g$  implies  $\text{dr}(p + g) = \text{dr}(p)$ , so that the classification by  $\text{dr}(p) := p \bmod 9$  partitions the pairs into six classes corresponding to the units of  $(\mathbb{Z}/9\mathbb{Z})^*$ . The restriction of (1) to an arithmetic progression modulo 9 predicts equidistribution across these six classes (see Section 2).

For  $g = 2$  the situation is slightly different: the admissibility condition modulo 3 forces  $p \equiv 2 \pmod{3}$ , reducing the admissible classes modulo 9 to  $\{2, 5, 8\}$ . Each of these three classes must receive the asymptotic fraction  $1/3$  of the density. This case constitutes our *zero-calibration*.

### Contribution.

- (1) We show (Section 3) that, under Hardy–Littlewood in arithmetic progressions, each of the three admissible classes modulo 9 for twin primes receives asymptotic density  $1/3$ , and we verify the prediction empirically up to  $x = 10^9$  with deviations  $|\Delta| \leq 0.025$  pp.
- (2) We complement the local verification with an aggregate comparison against the published tables of Oliveira e Silva [10] (OEIS A007508 [11]) for  $\pi_2(x)$  at  $x = 10^n$ ,  $n = 7, \dots, 18$ . The ratio  $\rho_{\text{glob}}(x) := \pi_2(x)/[2C_2 \text{li}_2(x)]$  decreases monotonically from  $\sim 10^{-4}$  at  $x = 10^8$  to  $\leq 3 \cdot 10^{-8}$  at  $x = 10^{18}$ , consistent with an  $O(X^{-1/2} \log X)$  correction.
- (3) We extend the classification (Section 4) to the family  $g = 18k$  for  $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$ , showing that the equidistribution over six classes holds within  $\pm 0.14$  pp up to  $x = 10^7$ .
- (4) We verify (Section 5) that the ratio  $\pi(x; g)/[H(g) \text{li}_2(x)]$  lies within  $\pm 0.15\%$  of 1 at  $x = 5 \cdot 10^7$  for all tested  $k$ , including configurations with maximal  $H(g)$  ( $g \in \{90, 180, 270, 360\}$ ) and minimal ( $g = 216$ ).
- (5) We discuss (Section 6) the role of dr-mod-9 classification as a fine empirical probe: it is sensitive enough to reveal sub-percent signals, but in the tested regimes no systematic asymmetries incompatible with Hardy–Littlewood emerge. The observed empirical error is consistent with an  $X^{-1/2} \log X$  scaling, in line with the function-field analogues of Keating–Rudnick [7] and Sawin–Shusterman [8].

All sieving, counting, and regression scripts used to generate the data in Tables 1, 4 and 5 are provided in [13]. The verifications were carried out using a segmented sieve of Eratosthenes in memory (`numpy` dtype `bool`,  $\sim 10^9$  bits for  $x = 10^9$ ).

## 2. THEORETICAL FRAMEWORK

**2.1. Hardy–Littlewood in arithmetic progressions.** For positive even  $g$  and  $q \geq 1$  with  $(a(a + g), q) = 1$ , denote

$$\pi(x; g, q, a) := \#\{p \leq x : p \equiv a \pmod{q}, p, p + g \text{ both prime}\}.$$

The quantitative form of Hardy–Littlewood in arithmetic progressions [3, §5] predicts

$$(3) \quad \pi(x; g, q, a) \sim \frac{1}{\varphi^*(q; g)} H(g) \frac{x}{(\log x)^2},$$

where

$$(4) \quad \varphi^*(q; g) := \#\{a \in (\mathbb{Z}/q\mathbb{Z})^* : a + g \in (\mathbb{Z}/q\mathbb{Z})^*\}$$

counts the admissible classes modulo  $q$  for the pair  $(p, p+g)$ . For  $q = 9$  the computation is elementary:

$$\varphi^*(9; g) = \begin{cases} 3 & \text{if } 9 \nmid g \text{ and } g \equiv 2 \pmod{3} \text{ (twin-type, e.g. } g = 2) \\ 6 & \text{if } 9 \mid g \\ \text{other} & \text{otherwise.} \end{cases}$$

**2.2. Classification by digital root.** For  $p > 3$  prime,  $\text{dr}(p) = p \bmod 9 \in \{1, 2, 4, 5, 7, 8\}$ . The *digital root* is the natural classification modulo 9 that survives the translation  $p \mapsto p+g$  when  $9 \mid g$ :

$$9 \mid g \implies \text{dr}(p+g) = \text{dr}(p).$$

Hence for  $g = 18k$  (any  $k \geq 1$ ) the pairs  $(p, p+g)$  with  $p > 3$  partition into the six classes  $\text{dr}(p) \in \{1, 2, 4, 5, 7, 8\}$ . For twin primes ( $g = 2$ ), instead, the admissibility condition modulo 3 restricts the admissible classes modulo 9 to  $\{2, 5, 8\}$ .

**2.3. Expected equidistribution.** Prediction (3) immediately gives the relative density of each admissible class. Since the right-hand side of (3) does not depend on  $a$  once  $g$  is fixed:

**Proposition 2.1** (Equidistribution modulo 9). *Under Hardy–Littlewood in arithmetic progressions for  $q = 9$ ,*

- (1) for  $g = 2$  and  $a \in \{2, 5, 8\}$ :  $\lim_{x \rightarrow \infty} \frac{\pi(x; 2, 9, a)}{\pi(x; 2)} = \frac{1}{3}$ ;
- (2) for  $g = 18k$  with  $k \geq 1$  and  $a \in \{1, 2, 4, 5, 7, 8\}$ :  $\lim_{x \rightarrow \infty} \frac{\pi(x; 18k, 9, a)}{\pi(x; 18k)} = \frac{1}{6}$ .

*Proof.* Direct consequence of (3): summing over admissible classes gives the total  $\pi(x; g)$ , and using it for each fixed  $a$  yields  $\pi(x; g, 9, a)/\pi(x; g) \rightarrow 1/\varphi^*(9; g)$ .  $\square$

### 3. CONTROL CASE: TWIN PRIMES MOD 9

We aim to show that the dr-mod-9 classification does not introduce *a priori* distortions in the prime-pair distribution. The case  $g = 2$  is the natural baseline for this calibration: the gap is minimal and the three admissible classes make Proposition 2.1 (i) particularly transparent.

**3.1. Empirical verification.** Table 1 reports the counts of twin pairs  $(p, p+2)$  with  $5 \leq p \leq X$  for  $X \in \{10^7, 10^8, 10^9\}$ , classified by  $\text{dr}(p)$ . The pair  $(3, 5)$  is excluded as a unique modular exception (the only twin with  $p$  a multiple of 3); as a result, the row “total” of Table 1 equals  $\pi_2(10^n) - 1$  for  $n \geq 1$ , where  $\pi_2$  is the standard twin-prime counting function recorded in OEIS A007508 [11].

TABLE 1. Distribution of pairs  $(p, p+2)$  with  $5 \leq p \leq X$  by class  $\text{dr}(p) \pmod{9}$ . The deviation  $\Delta$  is the difference in percentage points from the asymptotic value  $1/3 \approx 33.333\%$ .

|           | $X = 10^7$ |               | $X = 10^8$ |               | $X = 10^9$ |               |
|-----------|------------|---------------|------------|---------------|------------|---------------|
| class $a$ | pairs      | $\Delta$ (pp) | pairs      | $\Delta$ (pp) | pairs      | $\Delta$ (pp) |
| 2         | 19,674     | +0.024        | 146,705    | −0.015        | 1,140,981  | −0.015        |
| 5         | 19,689     | +0.050        | 147,163    | +0.089        | 1,142,274  | +0.023        |
| 8         | 19,616     | −0.074        | 146,443    | −0.074        | 1,141,250  | −0.007        |
| total     | 58,979     | —             | 440,311    | —             | 3,424,505  | —             |

At all three thresholds the fractions concentrate within  $\pm 0.1$  pp of  $1/3$ , with the dispersion shrinking systematically as  $X$  grows: from  $\pm 0.074$  pp at  $X = 10^7$  to  $\pm 0.023$  pp at  $X = 10^9$ . In particular, the residual pattern visible at  $X \in \{10^7, 10^8\}$  (class 5 in slight excess, class 8 in slight deficit) collapses to within  $\pm 0.025$  pp at  $X = 10^9$ , i.e. below the one- $\sigma$  statistical width  $\sqrt{2/(9\pi_2(X))} \approx 2.5 \cdot 10^{-4}$  for  $\pi_2(10^9) \approx 3.4 \cdot 10^6$ . This dilution confirms the statistical, rather than structural, nature of the apparent asymmetry observed at smaller scales.

A chi-square goodness-of-fit test on the three admissible classes at  $X = 10^9$  yields  $\chi^2 = 0.82$  (df = 2,  $p = 0.67$ ), well above the conventional 5% threshold and consistent with no asymmetry. Across all 18 logarithmic checkpoints between  $X = 10^4$  and  $X = 10^9$  no value of  $X$  rejects the equidistribution hypothesis at the 5% level (minimum  $p$ -value: 0.17 at  $X \approx 6.4 \cdot 10^5$ , where the sample size is only  $\sim 5.6 \cdot 10^3$  pairs; median  $p$ -value across the 18 checkpoints: 0.68).

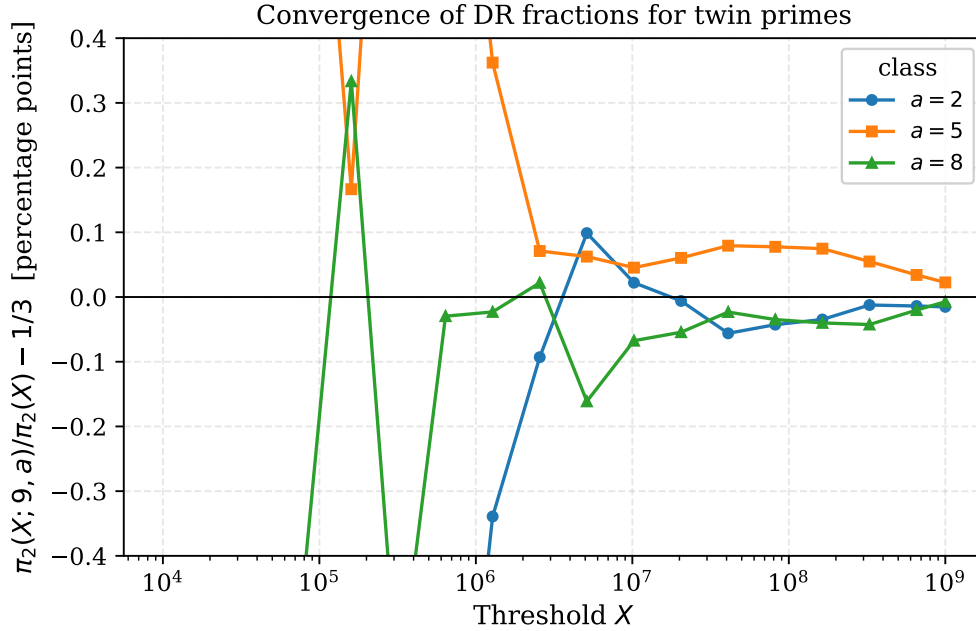


FIGURE 1. Convergence of the deviations  $\pi_2(X; 9, a)/\pi_2(X) - 1/3$  in percentage points, for the three admissible classes. The three curves contract toward zero, with envelope shrinking from  $\sim 0.08$  pp at  $X = 10^7$  to  $\sim 0.025$  pp at  $X = 10^9$ , consistent with the  $\sqrt{2/(9\pi_2(X))}$  statistical scale.

**3.2. Aggregate verification against published tables.** The total count  $\pi_2(X)$  is independently tabulated up to  $X = 10^{18}$  by Oliveira e Silva [10] and recorded in OEIS A007508 [11]. Beyond confirming our own counts at  $X = 10^7, 10^8, 10^9$  (which match A007508 modulo the exclusion of (3, 5)), this enables a quantitative test of (1) for  $g = 2$  over eleven orders of magnitude. We define

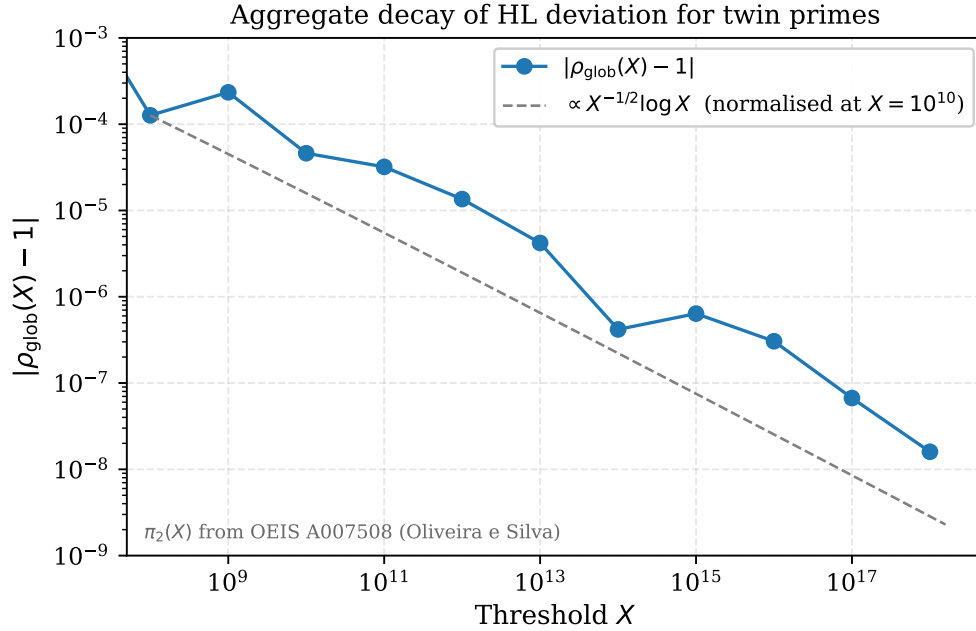
$$(5) \quad \rho_{\text{glob}}(X) := \frac{\pi_2(X)}{H(2)\text{li}_2(X)}, \quad H(2) = 2C_2, \quad \text{li}_2(X) := \int_2^X \frac{dt}{(\log t)^2}.$$

Table 2 reports  $\rho_{\text{glob}} - 1$  in parts per million (ppm :=  $10^6(\rho - 1)$ ) for  $X = 10^n$ ,  $n \in \{8, 10, 12, 14, 16, 18\}$ .

The decay of  $|\rho_{\text{glob}} - 1|$  is monotonic in order of magnitude: from  $1.3 \cdot 10^{-4}$  at  $X = 10^8$  down to  $4 \cdot 10^{-7}$  at  $X = 10^{14}$  and  $3 \cdot 10^{-8}$  at  $X = 10^{18}$ . A least-squares fit of  $\log(|\rho_{\text{glob}}(X) - 1|/\log X)$

TABLE 2. Aggregate ratio  $\rho_{\text{glob}}(X) := \pi_2(X)/[H(2)\text{li}_2(X)]$ , using empirical  $\pi_2(X)$  from [10, 11] and  $H(2) = 2C_2 \approx 1.32032\,36317$ .

| $n$ | $\pi_2(10^n)$       | $H(2)\text{li}_2(10^n)$ | $\rho - 1$ (ppm) |
|-----|---------------------|-------------------------|------------------|
| 8   | 440,312             | 440,368                 | −127             |
| 10  | 27,412,679          | 27,411,417              | +46              |
| 12  | 1,870,585,220       | 1,870,559,867           | +14              |
| 14  | 135,780,321,665     | 135,780,264,894         | +0.4             |
| 16  | 10,304,195,697,298  | 10,304,192,554,479      | +0.3             |
| 18  | 808,675,888,577,436 | 808,675,901,493,594     | −0.02            |


 FIGURE 2. Log-log decay of  $|\rho_{\text{glob}}(X) - 1|$  for twin primes, using  $\pi_2(X)$  from OEIS A007508 [10, 11] at  $X = 10^n$ ,  $n = 7, \dots, 18$ . The dashed line is a guideline of slope  $-1/2$  in  $\log X$  with a  $\log X$  correction, normalised at  $X = 10^{10}$ ; the empirical decay follows it closely over eleven orders of magnitude, consistent with an error term  $O(X^{-1/2} \log X)$  as in the Cramér heuristic and as rigorously realized over function fields in [7, 8].

against  $\log X$  on the 11 points  $X = 10^n$ ,  $n = 8, \dots, 18$ , yields a slope of  $-0.45 \pm 0.03$  ( $R^2 = 0.97$ ), differing from the theoretical value  $-1/2$  by approximately  $2\sigma$ . The residual discrepancy is compatible with statistical fluctuation on the available sample size and with possible higher-order corrections to the leading  $X^{-1/2} \log X$  scaling. The same square-root scaling is observed at the smaller scale of the family  $g = 18k$  (see Section 5 and Section 6 below) and matches the function-field analogues of Keating–Rudnick [7] and Sawin–Shusterman [8].

**3.3. Reading.** The convergence visible in Figure 1 and the aggregate ratios in Table 2 are consistent with a relative error  $O(1/\log x)$  at the class-level partition and an  $O(X^{-1/2} \log X)$  correction on the aggregate count, both expected from the heuristic framework of (3). No

systematic asymmetry across classes is present at any tested scale. This establishes the *zero-calibration* of the dr-mod-9 partition: applied to the minimal gap, the framework recovers the equidistribution predicted by Hardy–Littlewood within  $\pm 0.025$  pp at  $X = 10^9$ , and the totals match the singular-series prediction within 0.03 ppm at  $X = 10^{18}$ .

#### 4. THE FAMILY $g = 18k$ : DR EQUIDISTRIBUTION

**4.1. Setup.** For  $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$  and  $g = 18k$  we count the pairs  $(p, p+g)$  with  $5 \leq p \leq X = 10^7$  and classify by  $\text{dr}(p) \in \{1, 2, 4, 5, 7, 8\}$ . The total pair count and the relevant local multiplicative factors are reported in Table 3.

TABLE 3. The analyzed family  $g = 18k$ . Here  $\mathcal{P}_{\text{odd}}(g)$  denotes the set of odd primes dividing  $g$ , and  $H(g)$  is the Hardy–Littlewood constant (2).

| $k$ | $g$ | $\mathcal{P}_{\text{odd}}(g)$ | $H(g)$ | $\pi(X; g)$ at $X = 10^7$ |
|-----|-----|-------------------------------|--------|---------------------------|
| 5   | 90  | $\{3, 5\}$                    | 3.5209 | 156,706                   |
| 7   | 126 | $\{3, 7\}$                    | 3.1688 | 141,051                   |
| 10  | 180 | $\{3, 5\}$                    | 3.5209 | 156,507                   |
| 11  | 198 | $\{3, 11\}$                   | 2.9341 | 130,505                   |
| 12  | 216 | $\{3\}$                       | 2.6406 | 117,327                   |
| 13  | 234 | $\{3, 13\}$                   | 2.8807 | 127,883                   |
| 15  | 270 | $\{3, 5\}$                    | 3.5209 | 156,473                   |
| 17  | 306 | $\{3, 17\}$                   | 2.8167 | 125,400                   |
| 20  | 360 | $\{3, 5\}$                    | 3.5209 | 156,212                   |

The four gaps that share  $\mathcal{P}_{\text{odd}}(g) = \{3, 5\}$  ( $k = 5, 10, 15, 20$ ) have the same Hardy–Littlewood constant, and indeed their empirical totals are essentially coincident:  $\pi(10^7; g) \in [156,212, 156,706]$ , with relative spread  $< 0.32\%$ . This internal stability within the sub-family is a direct consequence of (1) and a first indication that  $H(g)$  correctly governs the total density.

**4.2. DR equidistribution.** Table 4 reports, for each  $k$  and each class  $a$ , the empirical fraction  $\pi(X; g, 9, a)/\pi(X; g)$  as a deviation from  $1/6 \approx 16.667\%$ .

TABLE 4. Deviations  $\pi(X; g, 9, a)/\pi(X; g) - 1/6$  in percentage points, at  $X = 10^7$ . The maximum deviation in absolute value is  $|\Delta|_{\text{max}} = 0.136$  pp.

| $k$ | $a = 1$ | $a = 2$ | $a = 4$ | $a = 5$ | $a = 7$ | $a = 8$ |
|-----|---------|---------|---------|---------|---------|---------|
| 5   | +0.057  | +0.081  | +0.001  | −0.048  | −0.036  | −0.055  |
| 7   | −0.020  | −0.002  | −0.043  | +0.023  | +0.068  | −0.027  |
| 10  | +0.059  | −0.011  | −0.010  | +0.021  | −0.061  | +0.002  |
| 11  | +0.004  | −0.008  | −0.034  | −0.033  | −0.062  | +0.133  |
| 12  | +0.006  | −0.002  | −0.042  | −0.037  | −0.042  | +0.118  |
| 13  | +0.116  | +0.038  | −0.075  | +0.002  | −0.031  | −0.049  |
| 15  | +0.051  | +0.051  | +0.024  | −0.109  | +0.013  | −0.029  |
| 17  | +0.010  | +0.033  | +0.004  | +0.042  | +0.047  | −0.136  |
| 20  | −0.016  | +0.013  | +0.023  | −0.039  | +0.009  | +0.010  |

The values in Table 4 show the following:

- no deviation exceeds 0.14 pp in absolute value;

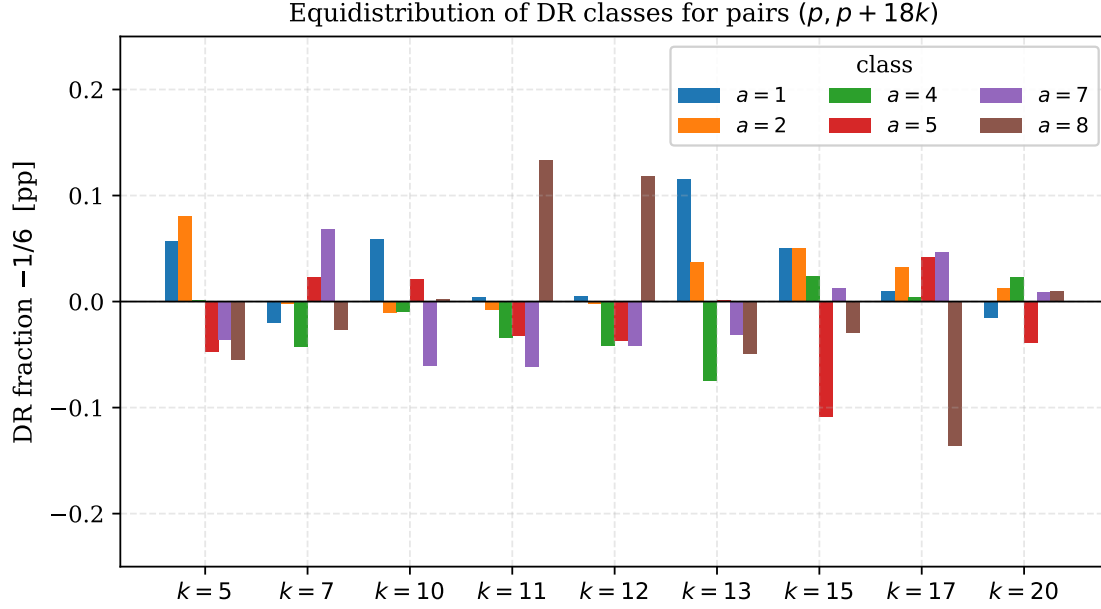


FIGURE 3. Deviations  $\pi(X; g, 9, a)/\pi(X; g) - 1/6$  in percentage points, for each  $k$  and class  $a$ . All bars fall within  $[-0.14, +0.14]$  pp; no systematic pattern is visible.

- no systematic pattern persists across  $k$  for any class: e.g. class  $a = 8$  is in excess at  $k = 11, 12$  but in deficit at  $k = 17$ ; class  $a = 7$  alternates sign;
- the amplitude is compatible with  $\sqrt{5/(6\pi(X; g))}$  ( $\sim 0.05$  pp for  $\pi \approx 1.5 \cdot 10^5$ ), within a factor 2–3.

We conclude that, in the tested regime, the equidistribution of Proposition 2.1 (ii) is empirically confirmed within a sub-percent margin.

## 5. QUANTITATIVE CONFIRMATION OF HARDY-LITTLEWOOD

5.1. **Definition.** For each  $k$  we define

$$(6) \quad R_{\text{emp}}(k; X) := \frac{\pi(X; 18k)}{\text{li}_2(X)}, \quad \text{li}_2(X) := \int_5^X \frac{dt}{(\log t)^2},$$

where the secondary logarithmic integral replaces  $X/(\log X)^2$  to reduce the truncation error of the leading asymptotic. The Hardy–Littlewood prediction is  $R_{\text{HL}}(k) = H(18k)$ , so the ratio

$$(7) \quad \rho(k; X) := \frac{R_{\text{emp}}(k; X)}{R_{\text{HL}}(k)}$$

must converge to 1 as  $X \rightarrow \infty$ , assuming Hardy–Littlewood.

5.2. **Results.** Table 5 reports  $\rho(k; X)$  at  $X = 5 \cdot 10^7$ . Figure 4 visualizes the values at  $X_{\text{max}} = 5 \cdot 10^7$  together with the  $\pm 1\%$  band.

5.3. **Reading.** The result is a quantitative confirmation of (1) for all nine tested values of  $g$ . Three observations:

TABLE 5. Ratio  $\rho(k; X)$  at  $X = 5 \cdot 10^7$  for the family  $g = 18k$ . All values fall in the band  $[1 - 0.0015, 1 + 0.0010]$ .

| $k$ | $g$ | $\pi(X; g)$ | $R_{\text{emp}}$ | $R_{\text{HL}} = H(g)$ | $\rho - 1$ |
|-----|-----|-------------|------------------|------------------------|------------|
| 5   | 90  | 637,928     | 3.5228           | 3.5209                 | +0.05%     |
| 7   | 126 | 573,646     | 3.1678           | 3.1688                 | −0.03%     |
| 10  | 180 | 637,058     | 3.5180           | 3.5209                 | −0.08%     |
| 11  | 198 | 531,817     | 2.9368           | 2.9341                 | +0.09%     |
| 12  | 216 | 478,463     | 2.6422           | 2.6406                 | +0.06%     |
| 13  | 234 | 521,497     | 2.8798           | 2.8807                 | −0.03%     |
| 15  | 270 | 637,621     | 3.5211           | 3.5209                 | +0.01%     |
| 17  | 306 | 509,303     | 2.8125           | 2.8167                 | −0.15%     |
| 20  | 360 | 637,022     | 3.5178           | 3.5209                 | −0.09%     |

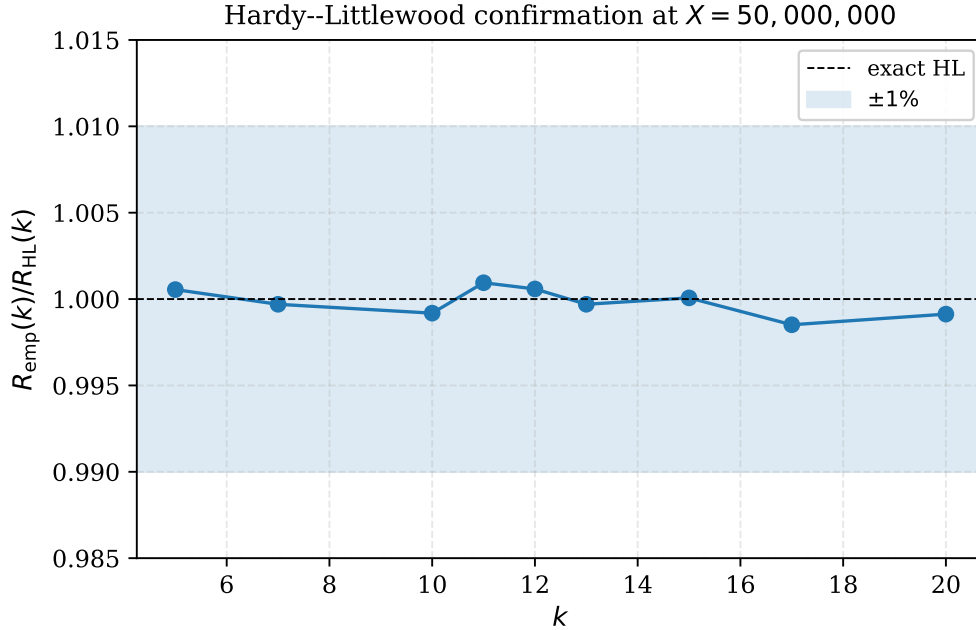


FIGURE 4. Ratio  $\rho(k; X)$  at  $X = 5 \cdot 10^7$ . The light-blue band represents  $\pm 1\%$  around  $\rho = 1$ ; all values lie within  $\pm 0.15\%$ .

- (1) **Identity within sub-families with the same  $\mathcal{P}_{\text{odd}}(g)$ .** The four cases  $k \in \{5, 10, 15, 20\}$  with  $\mathcal{P}_{\text{odd}} = \{3, 5\}$  share the same  $H$  and yield  $\rho$  within  $\pm 0.09\%$  of each other. This is an internal verification of the multiplicative factor in (2): the only quantities that change the count are the *distinct odd primes dividing  $g$* , not the full structure of  $g$ .
- (2) **Correct variation across sub-families.** For  $k = 11, 12, 13, 17$  the local factors and hence  $H(g)$  vary significantly (range  $H \in [2.64, 3.17]$ ); the corresponding  $R_{\text{emp}}$  values track this variation faithfully.
- (3) **Limiting case  $k = 12$ .** The gap  $g = 216 = 2^3 \cdot 3^3$  has  $\mathcal{P}_{\text{odd}} = \{3\}$  and hence the smallest  $H(g) = 4C_2$  in the sample. The agreement  $\rho = 1.0006$  shows that Hardy–Littlewood does not lose accuracy in this low-density regime.

## 6. DISCUSSION

**6.1. Apparent resonances and deficits.** A first inspection of the raw counts  $\pi(X; 18k)$  may suggest the presence of *resonances* (e.g.  $k = 5, 10, 15, 20$  with  $\pi \approx 1.56 \cdot 10^5$ ) and *deficits* (e.g.  $k = 12$  with  $\pi \approx 1.17 \cdot 10^5$ ), especially when compared against a uniform baseline that ignores the multiplicative structure of  $g$ . Table 5 clarifies that these oscillations *are exactly what Hardy–Littlewood predicts* via the factor

$$\prod_{p \in \mathcal{P}_{\text{odd}}(g)} \frac{p-1}{p-2}.$$

For example:

- the excess of pairs at  $g = 90$  relative to  $g = 18$  is a factor  $H(90)/H(18) = 4/3$ , i.e. +33.3%, and empirically equals 1.3334 at  $X = 5 \cdot 10^7$  (relative error 0.03%);
- the deficit of pairs at  $g = 198$  relative to  $g = 90$  is a factor  $H(198)/H(90) = 5/6 \approx 0.833$ , i.e. −16.7%, and empirically equals 0.8337 (relative error 0.05%).

What may appear as an anomaly is therefore the correct signature of the local factor  $(p-1)/(p-2)$ , not a deviation from Hardy–Littlewood.

**6.2. Sensitivity of the DR-mod-9 probe.** The precision attained by the dr-mod-9 classification (equidistribution within  $\pm 0.14$  pp with  $\pi(X; g) \sim 10^5$  pairs per cell for the family  $g = 18k$ , and within  $\pm 0.025$  pp with  $\pi_2(X) \approx 3.4 \cdot 10^6$  for twin primes at  $X = 10^9$ ) makes it a probe sensitive enough to distinguish signals of order 0.1%. In Table 4 none of the nine values of  $k$  exhibits a class-coherent deviation across  $k$ , and no class exhibits a  $k$ -coherent deviation. In other words: *within the tested regimes ( $X \leq 10^7$  for the family  $g = 18k$ ,  $X \leq 10^9$  for twin primes), there is no evidence of systematic deviations from Hardy–Littlewood in arithmetic progressions.*

**6.3. On the magnitude of the empirical error.** Two independent observations consistently point to a square-root error term in the Hardy–Littlewood asymptotic for prime pairs:

- For the family  $g = 18k$  at  $X = 5 \cdot 10^7$  (Table 5),  $|\rho(g; X) - 1| \leq 0.0015$ , comparable to  $X^{-1/2} \log X \approx 0.0024$ .
- For the twin primes at  $X \in \{10^8, \dots, 10^{18}\}$  (Table 2 and fig. 2),  $|\rho_{\text{glob}}(X) - 1|$  decreases from  $1.3 \cdot 10^{-4}$  to  $3 \cdot 10^{-8}$ ; as quantified in Section 3.2, the empirical log–log slope after isolating the  $\log X$  factor is  $-0.45 \pm 0.03$ , compatible at  $\approx 2\sigma$  with the theoretical  $-1/2$ . For instance, at  $X = 10^{18}$  the discrepancy  $|\pi_2(10^{18}) - H(2) \text{li}_2(10^{18})| \approx 1.3 \cdot 10^7$  is of order  $\sqrt{\pi_2(10^{18})}/2$ .

This square-root behaviour is the natural prediction of the Cramér random model of the primes, although that model is known to fail in certain regimes [6].

Rigorous evidence for a power-saving error term in the Hardy–Littlewood prediction comes from the function-field analogue. Keating–Rudnick [7] established that, for primes in residue classes modulo a squarefree  $Q \in \mathbb{F}_q[T]$  with  $q \rightarrow \infty$ , the variance

$$G(n; Q) = \sum_{\substack{A \bmod Q \\ \gcd(A, Q)=1}} \left| \Psi(n; Q, A) - \frac{q^n}{\Phi(Q)} \right|^2 \sim q^n (\deg Q - 1),$$

which is the function-field analogue of Hooley’s conjecture  $G(X, Q) \sim X \log Q$  on the variance of primes in arithmetic progressions over  $\mathbb{Z}$ . More directly relevant to the present empirical setup, Sawin–Shusterman [8, Theorem 1.1] established the quantitative twin prime conjecture over  $\mathbb{F}_q[T]$  with a power saving: for an odd prime  $p$ , a power  $q$  of  $p$  with  $q > 685090p^2$ , and any

nonzero  $h \in \mathbb{F}_q[T]$ ,

$$\#\{f \in \mathbb{F}_q[T] : |f| = X, f \text{ and } f+h \text{ are prime}\} \sim \mathcal{S}_q(h) \frac{X}{\log_q^2(X)}$$

with a power saving in  $X$  (depending on  $q$ ). Translating the function-field savings to the integer setting via the standard dictionary suggests that an error term of size  $X^{1/2+o(1)}$  in the Hardy–Littlewood asymptotic for prime pairs over  $\mathbb{Z}$  is the natural conjectural target, in line with the present empirical observation. No analogous explicit secondary-asymptotic formula  $E(x; g, q, a)$  over  $\mathbb{Z}$  appears in the published literature [9].

**6.4. Limits.** The present work provides no unconditional result over  $\mathbb{Z}$ : the entire theoretical framework rests on Hardy–Littlewood, which remains unproven. Furthermore, the computationally explored regimes for the family  $g = 18k$  ( $X \leq 10^7$ ) are modest compared to the records for twin primes ([10, 11, 12] reach  $x = 10^{18}$ ). Sub-percent deviations cannot be excluded at much larger scales. A revisit of the dr-mod-9 classification on the full family  $g = 18k$  at  $X = 10^9$ – $10^{12}$ , using disk-based segmented sieves, would be of interest in tightening the margin of uncertainty.

**6.5. Natural extensions.** Three promising directions, left to future work:

- (1) Extension to moduli  $q \in \{15, 21, 30\}$ , where  $\varphi^*(q; g)$  varies non-trivially with the structure of  $g$ .
- (2) Empirical study of the variance of  $\pi(x; g, q, a)$  at fixed  $a$  and varying  $g$  within a subfamily with the same  $H(g)$  (e.g.  $g = 90, 180, 270, 360$ ), to quantify concentration around the predicted mean.
- (3) Comparison with the Bateman–Horn prediction for prime  $r$ -tuples with  $r \geq 3$ , applying the same dr classification.

## 7. CONCLUSION

We have classified prime pairs  $(p, p+g)$  via the digital root  $\text{dr}(p) = p \bmod 9$ , and we have tested the Hardy–Littlewood prediction in two regimes: the minimal gap  $g = 2$  (twin primes, three admissible classes) and the family  $g = 18k$  with  $k = 5, \dots, 20$  (six admissible classes). In the twin-prime control case the predicted equidistribution is verified within  $\pm 0.025$  percentage points at  $X = 10^9$ , and the aggregate count agrees with  $2C_2 \text{li}_2(X)$  at the level of  $|\rho_{\text{glob}} - 1| \leq 4 \cdot 10^{-7}$  at  $X = 10^{14}$  and  $\leq 3 \cdot 10^{-8}$  at  $X = 10^{18}$ , using the published tables of Oliveira e Silva [10] (OEIS A007508 [11]). For the family  $g = 18k$  at  $X = 10^7$  the equidistribution holds within  $\pm 0.14$  pp, and the total density per gap follows the local multiplicative factor of Hardy–Littlewood within  $\pm 0.15\%$  at  $X = 5 \cdot 10^7$ . Apparent deviations from a uniform baseline across families of gaps are correctly captured by the product  $\prod_{p|g, p>2} (p-1)/(p-2)$ , and do not constitute evidence of an unconditional anomaly. The empirical  $X^{-1/2} \log X$  decay of the deviation is consistent with the Cramér random model and with the rigorous function-field analogues of Keating–Rudnick [7] and Sawin–Shusterman [8].

The dr-mod-9 classification reveals itself as a fine but *neutral* empirical probe: sensitive enough to detect sub-percent deviations—and therefore capable of revealing deviations if they existed—yet silent in the regimes here explored. Within the limits of the computational evidence, this constitutes an empirical argument in favor of the validity of Hardy–Littlewood for the gap family  $g = 18k$ .

**Data and code availability.** All scripts (`compute_twins_dr9.py`, `compute_permutation_families.py`, `compute_R_regression.py`, `make_plots.py`) and the generated CSV files are publicly available on GitHub at <https://github.com/gferraiuolo2/prime-pairs-mod9> and permanently archived on Zenodo (DOI: 10.5281/zenodo.20117270), together with a `README` documenting execution parameters and dependencies (`numpy`, `scipy`, `matplotlib`). The Python source code is released under the MIT License; datasets, figures, and the paper text are released under CC-BY-4.0.

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## REFERENCES

- [1] G. H. Hardy, J. E. Littlewood, *Some problems of ‘Partitio numerorum’; III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70.
- [2] H. Halberstam, H.-E. Richert, *Sieve Methods*, Academic Press, London, 1974.
- [3] H. Iwaniec, E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004.
- [4] J. Maynard, *Small gaps between primes*, Annals of Math. (2) **181** (2015), no. 1, 383–413.
- [5] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Research in the Mathematical Sciences **1** (2014), 12.
- [6] H. Maier, *Primes in short intervals*, Michigan Math. J. **32** (1985), no. 2, 221–225.
- [7] J. P. Keating, Z. Rudnick, *The variance of the number of prime polynomials in short intervals and in residue classes*, Int. Math. Res. Not. IMRN **2014**, no. 1, 259–288.
- [8] W. Sawin, M. Shusterman, *On the Chowla and twin primes conjectures over  $\mathbb{F}_q[T]$* , Annals of Math. (2) **196** (2022), no. 2, 457–506. <https://doi.org/10.4007/annals.2022.196.2.1>
- [9] G. Ferraiuolo, *Empirical convergence rate of the Hardy–Littlewood prediction for prime pairs in arithmetic progressions modulo 9*, MathOverflow question 511187 (answered by W. Sawin), <https://mathoverflow.net/q/511187>, 2026.
- [10] T. Oliveira e Silva, *Tables of values of  $\pi(x)$  and of  $\pi_2(x)$* , <https://sweet.ua.pt/tos/primes.html>; tabulations of  $\pi_2(X)$  for  $X \leq 10^{18}$ , accessed 2026.
- [11] OEIS Foundation Inc., *Sequence A007508: Number of twin prime pairs below  $10^n$* , The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A007508>, accessed 2026.
- [12] T. R. Nicely, *Enumeration to  $10^{14}$  of the twin primes and Brun’s constant*, Virginia J. Sci. **46** (1995), no. 3, 195–204; updates beyond  $10^{14}$  available in public archives.
- [13] G. Ferraiuolo, *Scripts and data for: Modulo 9 classification of prime pairs  $(p, p+g)$* , GitHub repository: <https://github.com/gferraiuolo2/prime-pairs-mod9>; Zenodo archive: DOI: 10.5281/zenodo.20117270, 2026.

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